

Database Management System			
Course Code:	CSE205	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	3U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Describe the fundamental elements of relational database management systems			
2. Explain the basic concepts of relational data model, entity-relationship model, relational database design, relational algebra and SQL.			
3. Design ER-models to represent simple database application scenarios			
4. Convert the ER-model to relational tables, populate relational database and formulate SQL queries on data.			
5. Improve the database design by normalization.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Students will have an understanding of database concepts and thorough knowledge of database software's.			
2. Students will be able to model an application's data requirements using ER diagrams			
3. Students will be able to write SQL commands to create tables and query data in a relational DBMS			
4. Students will be able to execute various advanced SQL queries related to transactions, concurrency			
5. Students will be able to explain the principle of transaction management design.			

UNIT I: DATABASE SYSTEM

Data base system vs. file system, view of data, data abstraction, instances and schemas, data models, ER model, relational model, DDL, DML, database access for applications programs, data base users and administrator, transaction management, data base system structure, storage manager, query processor, history of data base systems, data base design and ER diagrams, beyond ER design entities, attributes and entity sets, relationships and relationship sets, additional features of ER model, concept design with the ER model, and conceptual design for large enterprises.

UNIT II: RELATIONAL MODEL

Introduction to the relational model, integrity constraint over relations, enforcing integrity constraints, querying relational data, and logical data base design, destroying /altering tables and views. relational algebra and calculus: relational algebra, selection and projection set operations, renaming, joins, division.

UNIT III: BASIC SQL QUERY

Examples of basic SQL queries, nested queries, correlated nested queries set, comparison operators, aggregative operators, NULL values, comparison using null values, logical connectivity's, AND, OR and NOT, impact on SQL constructs, outer joins, disallowing NULL values, complex integrity constraints in SQL triggers and active data bases.

UNIT IV: SCHEMA REFINEMENT

Problems caused by redundancy, decompositions, problem related to decomposition, reasoning about FDS, FIRST, SECOND, THIRD normal form, BCNF, forth normal form, lossless join decomposition,

dependency preserving decomposition, schema refinement in data base design, multi valued dependencies.

UNIT V: OVERVIEW OF TRANSACTION MANAGEMENT

ACID properties, transactions and schedules, concurrent execution of transaction, lock based concurrency control, performance locking, and transaction support in SQL, crash recovery, concurrency control, Serializability and recoverability, lock management, lock conversions, dealing with deadlocks, specialized locking techniques, concurrency without locking, crash recovery: ARIES, log, other recovery related structures, the write, ahead log protocol, check pointing, recovering from a system crash, media recovery, other approaches and interaction with concurrency control.

Text Books:

1. Elmasri Navrate, Data Base Management System, Pearson Education, 2008.
2. Raghurama Krishnan, Johannes Gehrke, Data Base Management Systems, TMH, 3rd edition, 2008.
3. C. J. Date, Introduction to Database Systems, Pearson Education, 2009.
4. Silberschatz, Korth, Database System Concepts, McGraw hill, 5th edition, 2005.
5. Rob, Coronel & Thomson, Database Systems Design: Implementation and Management.